

TED (10)-4015
(REVISION-2010)

Reg. No.....
Signature.....

SIXTH SEMASTER DIPLOMA EXAMINATION IN CIVIL ENGINEERING

STRUCTURAL & IRRIGATION ENGINEERING DRAWING

MODEL QUESTION PAPER

(Time: 3 hrs)

(Maximum marks: 100)

- [Note: 1. Missing data if any can be suitably assumed.
2. Steel tables are permitted.
3. A2 size drawing sheet to be supplied.
4. Drawing shall be neat and fully dimensioned
5. Answer one full question from each unit]

Marks

MODULE - I

- I. Draw the following views of an R.C. beam as per the details given below:

- i) Longitudinal section
- ii) Cross – section at mid span
Cross – section at the face of the support

Details:

- | | |
|-----------------|--|
| Type of support | - Simply supported |
| Clear span | - 5 m |
| Wall thickness | - 300 mm |
| Overall depth | - 475 mm |
| Reinforcement | - tension bars, 3 Nos, 20 mm dia, Fe 415 bars.
Compression bars, 3 Nos, 16 mm dia, Fe 415 bars.
Stirrups at an average spacing of 150mm using 10mm dia,
Fe 415 bars |

(25)

OR

- II. The details of an RCC floor slab is given below:

- | | |
|--------------------------|--|
| Size of room | - 450 X 350 cm |
| Short span reinforcement | - 10 mm dia Fe 415 grade steel bars @ 120mm c/c,
Alternate bars bent up. |
| Long span reinforcement | - 10 mm dia Fe 415 grade steel bars @ 180 mm c/c,
Alternate bars bent up. |

Provide edge strip reinforcement and torsional reinforcement suitably:

- Draw: i) Sectional elevation along short span.
ii) Sectional elevation along long span.
iii) Details of torsional reinforcement.

(25)

MODULE - II

- III. A cantilever retaining wall has the following details:

- | | |
|------------------------|-------------|
| Filled earth level | - 100.000 m |
| Natural ground level | - 95.000 m |
| Top of base slab | - 94.00 m |
| Thickness of base slab | - 500 mm |
| Toe projection | - 1200 mm |
| Width of base slab | - 3300 mm |

Stem	- 300 mm at top 400 mm at bottom
Earth face vertical.	
Reinforcement details:	
Stem	- Inner face 16 mm dia Fe 415 grade steel @ 200 mm c/c. Alternate bars curtailed at 1/3 and 2/3 height from bottom.
Distributors	- 12 mm dia Fe 415 grade steel @ 250 mm c/c.
Outer face (vertical)	- 12 mm dia Fe 415 grade steel @ 250 mm c/c.
Distributors	- 10 mm dia bars Fe 415 @ 300 mm c/c.
Heel – Main bars	- 16 mm dia Fe 415 bars @ 180 mm c/c.
Distributors	- 12 mm dia Fe 415 bars @ 200 mm c/c.
Toe – Main bars	- 16 mm dia Fe 415 bars @ 150 mm c/c.
Distributors	- 12 mm dia Fe 415 bars @ 200 mm c/c.

Draw the following views:

i) C/S of the retaining wall.

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OR

IV Draw the sectional elevation and plan of a dog legged stair case with the following details:

Room size	- 6.0 m x 2.5 m
Floor height	-3.6 m
Slab thickness	-125 mm
Wall thickness	-300 mm
Width of landing	-1000 mm
Rise	-150 mm
Tread	-250 mm

Details of reinforcements:

Main reinforcement	-10 mm dia Fe 415 bars @ 150 mm c/c . Alternate Bars bent up.
Distributors	-8mm dia Fe 415 bars @ 180 mm c/c.

(25)

MODULE - III

V. An ISHB 350 x 250 @ 67.4 Kg/m is fixed to the bearing plate of 700 x 400 x 20 mm. The steel stanchion and bearing plate are connected by cleat angles ISA 100 x 100 x 10 mm on either sides of the flanges and on the web the same cleat angles are for both sides. Anchor bolt 20 mm dia, 4 Nos are used for connecting base plate and foundation footing. Draw front view, sectional plan of the gusseted base. Assume all connections are of riveted joints.

OR

VI Draw the front and side view of a cross beam connection with the following details:

Main beam	: ISLB 500 @ 75 kg/m
Secondary beams	: ISLM 300 @ 37.7 kg/m
Main beam cover plate	: 250 mm x 10 mm
Web cleat	: 2 ISA 90 x 90 x 8 mm
Flange angle	: ISA 100 x 75 x 8 mm

Rivets 3 Nos, 16 mm dia for web and flange cleats. Assume missing data if any.

(25)

MODULE – IV

VII. Draw the longitudinal section of a canal drop, along the center line of a canal, with the following Details:

U/S Bed level	:+ 11.50 m	FSL (u/s)	: + 12.50 m
Bund level	: +13.50 m	D/S bed level	:+ 11.00 m
Width of notch pier	: 0.65 m	Drop wall top	: 11.00 m
Drop wall bottom	:+ 8.60 m	Drop wall top width	: 1.0 m
Drop wall bottom width	: 2.60 m	Solid apron	: 1.0 m thick
Apron top level	:+ 8.60 m	FSL (d/s)	: +10.50 m
Tank bund level below drop		+11.50 m	

Necessary revetment and pitching are provided.

(25)

OR

VIII. Draw the longitudinal section of a surplus weir with the following details:

Hydraulic details – Bund top level	: +10.50
Maximum water level	: +9.00
Full tank level	: +8.40
Apron bottom	: +3.80
Tail channel	: +5.00

Side slope – U/S side – 1: 5: 1, D/S side – 2:1, Bund top width – 1.5, Splayed wing wall in RR Masonry in cement mortar 1:6, 50cm top and 200cm at bottom with side slope 1:8. Cement concrete 1:2:4 core wall – 90cm at top, 240cm at bottom, top level +8.40m. Thickness of apron – 50cm, Dam stones 20 x 20 x 60cm at 100cm c/c. Length of core wall is 300cm. Provide suitable cut off walls and other details.

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